UNIVERSITY OF CANBERRA
AUSTRALIA'S CAPITAL UNIVERSITY

Probing Preferences between Six Designs of Interactive Sonifications for Recreational Sports, Health and Fitness



Barrass, S.[1], Schaffert, N.[2] & Barrass, T. [3]

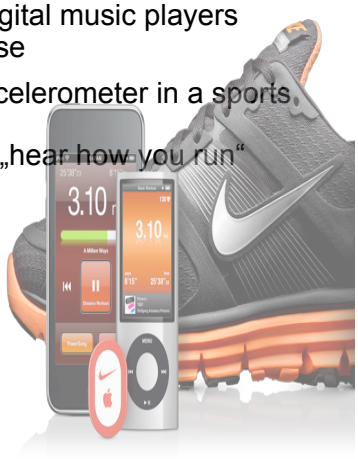
[1] University of Canberra, Australia
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[3] Sweatsonics, Australia



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Motivation

- 1980's Dance Aerobics made music common in Gyms
- People now wear headphones at the gym & jogging
- Music enhances endurance & enjoyment in sports
- Small, robust digital music players allow outdoor use
- Nike+iPod – accelerometer in a sports shoe
  selects songs - „hear how you run"



Questions arising

- How to design **function** *and* **aesthetics** for sonifications?
- Does the usage in a sporting activity change the appreciation and enjoyment of a sonification?
- Do different sonifications induce or support different kinds of activities?

Goals…

Introducing a *technology probe* methodology to
investigate sonifications in real-world sporting contexts...

Cycling, Jogging, Aerobics, Kayaking, Aikido, Walking, ...

A platform for rapid iterative design, prototyping and
trialling of interactive sonifications with feedback from
athletes physically engaged in the sport.

Recorded data and sound files can be used
in a post process as well.



Apparatus: Sweatsonic

sweatsonics – a 'technology probe' for interactive sonification



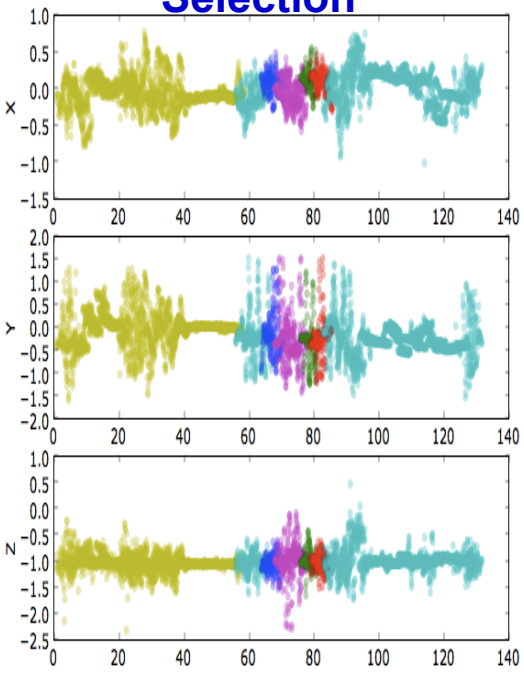
Selection *sonification*	
red	*algorithmic music*
yellow	*Sinification*
green	*weather metaphor*
cyan	*formants*
blue	*musicification*
magenta	*stream-based*

- Apple iPod touch with onboard 3-axis acceleration sensor (100 Hz)
  - Records timing of selections of different sonifications
  - audio output can also be recorded for later playback

Pilot study - interview and trial session

- Participants (*N*=15) = experts in Human Communication Science; males and females 20-60 years
- Pre-Interview about sonification in sport, use of music during exercises and introduction
- Probe was fitted to the upper arm and the sound information was transmitted via headphones
- Free trial-session with the probe as long as they wanted
- Post-session interview for overall impressions, preferences, and suggestions of applications and improvements

Acceleration coloured by Selection



Function and Aesthetics

Function in rowing

- audibly differentiated intensity steps and defined sections in the rowing cycle

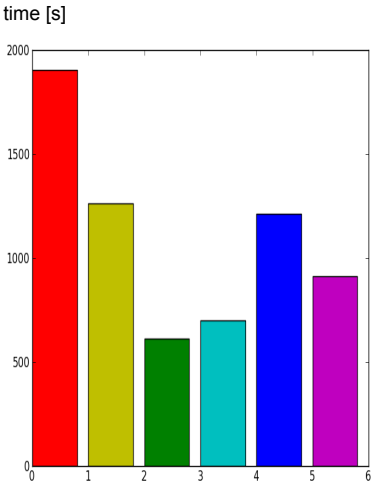
Aesthetics

- Sinification : function = aesthetics
- MIDIfication : does timbre matter ?
- Soundscape and Learning
- Metaphor, Context and Sound Design
- Musical Prototype
- Stream-based sonification - psychoacoustics

Device presentation



Results



- General pattern of preferences for *algorithmic* music, followed by *sinification* and *musification* across all 15 sessions
- But: different preferences among the participants: musical sounding sonification vs. informational sonic feedback

Results

- Participants differed in preference:
  - more conventional listening experience of a musical sounding sonification
  - more distinctly informational sonic feedback (like elite rowers)
- Preferences interpreted from the data corresponded with the most preferred sonifications in the post-interview

Conclusion

- The sweatsonics Technology Probe investigated preferences between six different interactive sonifications in recreational sporting activities
- The probe methodology allows experiments to be conducted in authentic outdoor physical activities rather than inside on gym equipment.
- Indications for future studies:
  - Sonification with athletes in a range of sports at different levels of competitiveness and in different physical and acoustic contexts



get Sweaty too!

download the sweatsonics probe

stephenbarrass.wordpress.com